┌Def. Let F be a field. Define the General Linear Group as;

$$GL_n(F) \overset{\text{def}}{=} \{A \in M_{n\times n}(F) \mid \det A \neq 0\}$$

$I_n \in GL_n(F)$, $\det A, \det B \neq 0 \Rightarrow \det AB \neq 0$, $\det A \neq 0 \Rightarrow \det A^{-1} \neq 0$, exists
Thus it is group.

┌Def. Define the Special Linear Group as; $SL_n(F) \overset{\text{def}}{=} \{A \in M_{n\times n}(F) \mid \det A = 1\}$

┌Lemma. Let $\mathbb{F}_q$ be a finite field of order q. Then,
$GL_n(\mathbb{F}_q)$ has order $(q^n-1)(q^n-q^1)\cdots(q^n-q^{n-1})$
$SL_n(\mathbb{F}_q)$ has order $\frac{1}{q-1}|GL_n(\mathbb{F}_q)|$

Proof. Clearly, $|M_{n\times n}(\mathbb{F}_q)| = q^{n^2}$.
Choice the first Column as non-zero, there are q^n-1 possibilities,
And, the second Column all but a multiple of the first, then q^2-q possibilities.
Successively, choose the third as all but not linear combination of first and second,
then there are q^n-q^2 (More specifically, q^2 is the number of cases of $a_1u_1+a_2u_2$).
Inductively, we obtain that $|GL_n(\mathbb{F}_p)| = \prod_{k=0}^{n-1}(q^n-q^k)$

Meanwhile, observe that

$$\varphi : GL_n(\mathbb{F}_q) \to \mathbb{F}_q^\times ; A \mapsto \det A$$

is well-defined (because $\det A$ is obtained by operations on $\mathbb{F}_q$), onto ($\det\begin{pmatrix}1 & & 0\\ & \ddots & \\ 0 & & a\end{pmatrix} = a$)
and Homomorphism ($\det AB = \det A \cdot \det B$) with $\ker\varphi = SL_n(\mathbb{F}_q)$. Since
$GL_n(\mathbb{F}_q)/\ker\varphi \cong \mathbb{F}_q^\times$, we obtain $|\ker\varphi| = |SL_n(\mathbb{F}_q)| = \frac{1}{q-1}\cdot|GL_n(\mathbb{F}_q)|$.

$(p-1)^2(p+1)$

「Problem.

1). Find order of $GL_2(\mathbb{F}_p)$, where p is the prime number.

2). Determine $P = \left\{ \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \middle| \, a \in \mathbb{F}_p \right\}$ is Sylow p-subgroup of $GL_2(\mathbb{F}_p)$.

3). Find order of $N_{GL_2(\mathbb{F}_p)}(P)$ and $\# Syl_p(GL_2(\mathbb{F}_p))$.

Proof. $A \in GL_2(\mathbb{F}_p) \Leftrightarrow \det A \neq 0$.

The number of possible vector in the first row in A is p^2-1, except zero vector.

For chosen first row, the possible number of second row is p^2-p, except scalar multiplication of the first row. So,

$$|GL_2(\mathbb{F}_p)| = (p^2-1)(p^2-p) = p\cdot(p-1)^2(p+1) = p(p^3-p^2-p+1)$$

Clearly P has order of p, and $p \nmid p^3-p^2-p+1$, so $P \in Syl_p(G)$.

Find $N_G(P)$. Since for any $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in GL_2$, where $ac-b \neq 0$,

$$\begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} \cdot \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \cdot \frac{1}{ac}$$

$$= \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} \begin{pmatrix} a & b+ck \\ 0 & c \end{pmatrix} \cdot \frac{1}{ac}$$

$$= \begin{pmatrix} ac & cb+c^2k-bc \\ 0 & ac \end{pmatrix} \cdot \frac{1}{ac}$$

$$= \begin{pmatrix} 1 & cka^{-1} \\ 0 & 1 \end{pmatrix} \in P,$$ therefore $|N_G(P)| = (p-1)^2 \cdot p$.

So, $n_p = [GL_2 : N_{GL_2}(P)] = p+1$. □

Def. The Heisenberg group over F defined by

$$H(F) \overset{\text{def}}{=} \left\{ \begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix} \;\middle|\; a_i \in F \right\} \leq SL_3(F).$$

Prop. $Z(H(F)) \cong (F, +)$

Proof observe that

$$\underbrace{\begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix}}_{A} \cdot \underbrace{\begin{pmatrix} 1 & b_1 & b_2 \\ 0 & 1 & b_3 \\ 0 & 0 & 1 \end{pmatrix}}_{=B} = \begin{pmatrix} 1 & a_1+b_1 & a_2+b_2+a_1b_3 \\ 0 & 1 & a_3+b_3 \\ 0 & 0 & 1 \end{pmatrix}$$

Therefore,

$$\begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a_1 & -(a_2-a_1a_3) \\ 0 & 1 & -a_3 \\ 0 & 0 & 1 \end{pmatrix}$$

And, $AB = BA \iff a_1b_3 = a_3b_1$. This implies

$$\begin{aligned} A \in Z(H(F)) &\iff \forall b_1, b_3 \in F,\ a_1b_3 = a_3b_1 \\ &\iff a_1 = a_3 = 0. \end{aligned}$$

Therefore,

$$Z(H(F)) = \left\{ \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \;\middle|\; a \in F \right\}.$$

Now, $F \to Z(H(F)) : a \mapsto \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is isomorphism, by

$$\begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & a+b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$\begin{array}{ll} GL_3(\mathbb{F}_q) & (q^3-1)(q^3-q)(q^3-q^2) \\ \quad | & \\ SL_3(\mathbb{F}_q) & (q^3-1)(q^3-q)q^2 \\ \quad | & \\ H(\mathbb{F}_q) & q^3 \\ \quad | & \\ Z(H(\mathbb{F}_q)) \cong (\mathbb{F}_q, +) & q \\ \quad | & \\ 1. & \end{array}$$